

Efficient Operation of Integrated Electrical-Water System for Wind Power Accommodation

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Abstract—This article investigates the operation of an integrated electrical-water system for wind power accommodation. The efficient operation can be considered as how well to handle the binary variables in a water distribution system (WDS) model. The optimization problem of WDS is NP-hard, and it often takes hours to find the optimal solution. Since wind power varies in minutes, long computation time may hinder the WDS from being fully used by power system to deal with the wind power fluctuation and uncertainties. This article develops a three-step optimization model to answer the challenges of the WDS optimization problem. The solution hinges on the calculation of the binary variable values in advance. The first step is to construct a linear power-WDS co-optimization model to find the optimal electrical demand of WDS. The second step develops a tank flow allocation based method and a greedy-based algorithm to determine the binary variable values of WDS. This allows the WDS optimization model to be efficiently solved by commercial solvers in the third step. The proposed three-step model converts the NP-hard optimization problem of WDS into a polynomial-time problem, which can be solved quickly. This allows the wind power system to leverage the energy flexibility provided by WDS in the real-time operation. Results of case study show that, compared with existing methods, the proposed method can reduce the wind power curtailment and the operation cost by about 55% and 53%. More importantly, the solution time is reduced dramatically from hours to seconds, suitable for practical application.

Index Terms—Binary variable, efficient operation, integrated electrical-water system, power variation range, renewable energy.

NOMENCLATURE

Most of the notations used throughout this article are defined below. Others are defined following their first appearances, as needed.

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Sets

Ω^T / Ω^W

Set of conventional generating units/wind farms.

Ω^L

Set of blocks in the piecewise function of the conventional generating unit.

Λ^E / Ψ^E

Set of electrical buses/lines.

$\Omega_n^T / \Omega_n^W / \Omega_n^{\text{WDS}}$

Set of units/wind farms/WDSs that connected to bus n .

Λ_n^E

Set of electrical buses connected to bus n .

$\Delta / \Delta_P / \Delta_V$

Set of water pipes/pumps/valves.

Ω_m

Set of tanks connected to node m .

$\varpi / \varpi_T / \Omega^{\text{WDS}}$

Set of water nodes/tanks/WDS.

Parameters

A_m^{tank}

Surface area of tank m (m^2).

$a_{P_m} / b_{P_m} / c_{P_m}$

Coefficients of the head-flow equation of pump m ($\text{m}^5 / \text{m}^2 / \text{m}$).

$B_{n,m}$

Susceptance of electrical line n - m (S).

c_{mn}

Loss coefficient of pipe m - n (m^{-5}).

$dw_{J_k}^t$

Demand of water node k at time t (m^3).

$d_{n,t}^{\text{er}}$

Forecasted demand of electrical load n at time t (MW).

$h_{J_m}^{\min}$

Minimum head of water node m (m).

$\kappa_1 / \kappa_2 / \kappa_3$

Coefficients of the power-flow equation of pump ($\text{m}^{-9} \times \text{MW} / \text{m}^{-6} \times \text{MW} / \text{m}^{-3} \times \text{MW}$).

$K_{i,l}$

Slope of block l of the piecewise function of unit i .

L

Number of piecewise linear cost function segments.

M'

A very large positive constant.

N_h^{WP}

Number of pumps in WDS h .

$N_{\text{WDS}} / N_W / N_{EB}$

Number of WDSs/wind farms/electrical loads.

N_g

Number of conventional units.

$P_{w,t}^{\text{wr}}$

Forecasted output of wind farm w at time t (MW).

$PF_{n,m}$

Capacity limit of electrical transmission line n - m (MVA).

$\hat{P}_{h,t_p}^{\text{WDS}}$

Optimal electrical demand of WDS at time t_p (MW).

$P_i^{\text{gmin}} / P_i^{\text{gmax}}$

Minimum/maximum output of unit i (MW).

$\hat{P}_{i,t_p-1}^{\text{g}}$

Optimized output of unit i at time $t_p - 1$ (MW).

$q_{mn}^{\min} / q_{mn}^{\max}$

Minimum/maximum volume of pipe m - n (m^3).

$qv_{mn}^{\min}/qv_{mn}^{\max}$	Minimum/maximum pipe flow when valve m - n on (m^3).
$q_{J_m}^r$	Maximum amount of flows that can be transported in or out of the tank m by the inlet and outlet pipes in a 15-min scale (m^3).
Ramp_i	Ramp power of unit i in a unit time (MW).
T	Number of time steps.
$\hat{\omega}_{P_m}$	Maximum allowable relative speed of the pump m .
$W_{J_m}^{\min}/W_{J_m}^{\max}$	Minimum/maximum water storage of tank m (m^3).
$\hat{x}_{i,t}$	Commitment status of the unit.
λ/μ	Penalty factor for wind power/electrical demand curtailment.
λprice_t	Electricity price at time t (\$/MWh).
Variables	
$h_{J_m}^t/h_{J_n}^t$	Head pressure of water node m/n at time t (m).
$\hat{h}_{P_m}^t$	Pressure head gain of pump m at time t (m).
$P_{i,t}^{WP}$	Electricity consumption of pump i at time t (MW).
$P_{h,t}^{\text{WDS}}$	Electricity consumption of WDS h at time t (MW).
$P_{w,t}^{\text{wc}}/P_{n,t}^{\text{ec}}$	Amount of wind power curtailment in the wind farm w /electrical demand curtailment at node n at time t (MW).
$P_{i,t,l}^g$	Power produced in block l of the piecewise linear function of the unit i at time t (MW).
$P_{i,t}^g$	Output of the unit i at time t (MW).
$q_{mn}^t/q_{P_m}^t$	Flow in pipe m - n /pump m at time t (m^3).
$q_{J_m,t}^{\text{in}}/q_{J_m,t}^{\text{out}}$	Water that flows into/out of tank m at time t (m^3).
R_{mn}^t	Head loss along the pipe m - n equipped with a valve at time t (m).
$\text{sign}(q_{mn}^t)$	Sign function which denotes the direction of flow in the pipe. It equals to 1 if $q_{mn}^t > 0$, zero when $q_{mn}^t = 0$, and -1 when $q_{mn}^t < 0$.
$W_{J_m,t}$	Volume of the stored water in tank m at time t (m^3).
$\omega_{P_m}^t$	Ratio of actual speed to the nominal speed of the pump m at time t .
z_{mn}^t	Binary variable indicating the ON/OFF status of the valve m - n at time t (equals 1 if on and 0 otherwise).
$\theta_{n,t}/\theta_{\text{ref},t}$	Phase angle of electrical node n /reference node at time t (rad).

I. INTRODUCTION

RENEWABLE energy has been widely used to build eco-friendly power system, and is estimated to meet 40% of the global electricity consumption by 2030 [1]. Renewable energy such as wind power, however, is uncertain and highly volatile compared with the power generated by conventional thermal generating units.

Various models have been developed for wind power forecasting and uncertainty quantification, such as the rough autoencoder [2], the deep belief network [3], the interval probability distribution learning [4], and the convolutional graph autoencoder [5]. These models can extract quantitative features from historical data of wind power output considering the uncertainties of data. Although accurate wind power forecasting can be obtained by use of the above models, high penetration of wind power has brought great challenges to the balance of supply and demand in the power system [6]. It is in urgent need of finding effective means for efficient operation of the wind power system.

Water distribution system (WDS) has appeared as a promising answer to the challenge mentioned previously. Several reports and pilot programs have shown that the integration of WDS with the power system could balance the wind power variations and reduce the wind power curtailment [7], [8], [9], [10]. WDS accounts for 5% of the annual total electricity consumption in China, and up to 7.1% in California, USA [11], [12]. The electrical demand of WDS can be adjusted flexibly by coordinating the operation of water pumps and tanks in the WDS. The water pump enables the increase of electricity consumption during the period of high wind power output and the decrease of electrical demand when the wind is scarce. The water tank is used to maintain the water balance in the WDS. More wind power can be incorporated into the power system through the optimization of WDS.

The components of WDS can be formulated mathematically and the cooperative operation model of the WDS and power system has been developed [13], [14], [15], [16]. The energy flexibility of WDS was leveraged to minimize the overall operation cost [17], [18]. The optimal bidding strategy for WDS to participate in the electricity market was also proposed [19], [20], [21]. The optimization model is generally formulated as a mixed-integer nonlinear programming (MINLP) problem [14], [15]. It consists of the binary variables and the continuous variables. The MINLP problem is NP-hard and unable to obtain the solutions quickly using the commercial solvers such as Gurobi [22]. Two factors affect the efficiency of solving the MINLP model of WDS [23]. One is the nonlinear terms in the hydraulic equations of the water pipe and pump [14]. Another factor is the binary variables describing the direction of flow in the pipe and the ON/OFF status of the pump and the valve [7].

Many efforts have been made to tackle the nonlinear terms in the WDS model with some success, such as the piecewise linear approximation [24], the second-order cone relaxation [25], the convex approximation [26], and the quasi-convex hull relaxation [27]. These methods transform the original MINLP model into a mixed-integer second-order cone programming model or a mixed-integer linear programming (MILP) model.

The binary variable concern in the MINLP model, however, remains unresolved. The quasi-convex hull relaxation method in [27] could remove the binary variables denoting the direction of flow in the pipe from the model, but not the binary variables indicating the ON/OFF status of the pump and the valve. In addition, the quasi-convex hull is not an exact representation of the original pipe model and may result in the water flow solution infeasible.

The number of binary variables increases drastically as the size of WDS expands. They split the feasible region of the optimization model into many discrete spaces and bring hefty computational expense. As the number of binary variables expands, the computation time for solving the MINLP model increases faster than exponential [23]. It usually takes hours to solve the optimization problem of a WDS [7]. The computation time becomes prohibitive when the size of system becomes very large [14]. Since the output of wind power varies rapidly, the optimization model of WDS should be solved in minutes to cope with the wind power uncertainties. The very long calculation time caused by the binary variables hinders the WDS from being fully used to optimize the wind power system operation. This makes the immediate response to the real-time variations of wind power impossible and poor handling of wind power uncertainties. As a result, the amount of wind power curtailment increases greatly, causing huge economic loss.

To the best of our knowledge, not many researches regarding the binary variables have been reported. The existing methods generally consider the binary variables and continuous variables in the WDS optimization. This complicates the solution finding [24], [25], [26], [27]. In this article, these variables will be treated separately. The values of binary variables are calculated in advance. With binary variables known, the optimization problem only consists of the continuous variables, which can be solved efficiently by commercial solvers.

This article proposes a three-step optimization approach for efficient operation of the integrated electrical-water system. The efficiency refers to lower wind power curtailment and shorter computation time. In specific, the time to obtain the solution is significantly reduced from hours to seconds. To achieve the efficient operation goal, the first step proposes the power-tank linear (PTL) constraints to describe the maximum variation range of the WDS power consumption, and develops a linear power-WDS co-optimization (LPWCO) model using these constraints. The LPWCO model is used to obtain the optimal electricity consumption profile of WDS. The second step proposes a greedy-based algorithm to determine the values of binary variables, ensuring that the optimized electrical demand of WDS in the first step is satisfied as much as possible. A tank flow allocation based method is also developed to reduce the computation cost of finding water flow solutions. The third step solves the WDS optimization model to obtain the optimal water flow solutions using the binary variable values in step 2.

The main contributions of this article can be summarized as follows.

- 1) Develop a three-step cooperative approach to determine the binary variable values of WDS in advance so as to facilitate efficient operation of the integrated electrical-water system.
- 2) Provide PTL constraints to describe the maximum variation range of the electricity consumption profile of WDS.
- 3) Propose a tank flow allocation based method to obtain the water flow solutions efficiently.

The rest of this article is organized as follows. The proposed optimization approach is described in Section II. It is composed

of three steps. Section III presents the first step and Section IV describes the other two steps. Case studies are given in Section V to verify the advantages of the proposed three-step optimization approach. Finally, Section VI concludes this article.

II. PROBLEM FORMULATION

The optimization model of WDS is composed of many binary variables and nonlinear constraints. This makes the WDS model a MINLP problem, which is generally NP-hard and often takes hours to find the optimal solution [7]. The constraints of WDS optimization model in this article can be formulated by equations described from [8]

$$h_{J_m}^t - h_{J_n}^t = c_{mn} \text{sign}(q_{mn}^t) (q_{mn}^t)^2 \quad \forall (J_m, J_n) \in \Delta \quad (1)$$

$$q_{mn}^{\min} \leq q_{mn}^t \leq q_{mn}^{\max} \quad \forall (J_m, J_n) \in \Delta \quad (2)$$

$$\text{Pump on : } \hat{h}_{P_m}^t = a_{P_m} (q_{P_m}^t)^2 + b_{P_m} q_{P_m}^t \omega_{P_m}^t + c_{P_m} (\omega_{P_m}^t)^2 \quad \forall P_m \in \Delta_P \quad (3)$$

$$\text{Pump on : } 0 \leq \omega_{P_m}^t \leq \hat{\omega}_{P_m} \quad \forall P_m \in \Delta_P \quad (4)$$

$$\text{Pump off : } q_{P_m}^t = 0 \quad \forall P_m \in \Delta_P \quad (5)$$

$$-M'(1 - z_{mn}^t) \leq h_{J_n}^t - h_{J_m}^t + R_{mn}^t \leq M'(1 - z_{mn}^t) \quad \forall (J_m, J_n) \in \Delta_V \quad (6)$$

$$R_{mn}^t \geq 0 \quad \forall (J_m, J_n) \in \Delta_V \quad (7)$$

$$qv_{mn}^{\min} z_{mn}^t \leq q_{mn}^t \leq qv_{mn}^{\max} z_{mn}^t \quad \forall (J_m, J_n) \in \Delta_V \quad (8)$$

$$dw_{J_k}^t = \sum_{J_n: (J_n, J_k) \in \Delta} q_{nk}^t - \sum_{J_m: (J_k, J_m) \in \Delta} q_{km}^t \quad \forall J_k \in \varpi \quad (9)$$

$$h_{J_m}^t \geq h_{J_m}^{\min} \quad \forall J_m \in \varpi \quad (10)$$

$$W_{J_m, t} = h_{J_m}^t A_{J_m}^{\text{tank}} \quad \forall J_m \in \varpi_T \quad (11)$$

$$-q_{J_m}^r \leq q_{J_m, t}^{\text{in}} + q_{J_m, t}^{\text{out}} \leq q_{J_m}^r \quad \forall J_m \in \varpi_T \quad (12)$$

$$W_{J_m, t+1} = W_{J_m, t} + q_{J_m, t}^{\text{in}} + q_{J_m, t}^{\text{out}} \quad \forall J_m \in \varpi_T \quad (13)$$

$$W_{J_m}^{\min} \leq W_{J_m, t} \leq W_{J_m}^{\max} \quad \forall J_m \in \varpi_T \quad (14)$$

$$P_{i, t}^{\text{WDP}} = \kappa_1 (q_{P_i}^t)^3 + \kappa_2 (q_{P_i}^t)^2 + \kappa_3 q_{P_i}^t \quad \forall P_i \in \Delta_P \quad (15)$$

$$P_{h, t}^{\text{WDS}} = \sum_{i=1}^{N_h^{\text{WDP}}} P_{i, t}^{\text{WDP}} \quad \forall h \in \Omega^{\text{WDS}}. \quad (16)$$

Constraint (1) describes the relationship between the head loss and the flow in a pipe using the Darcy–Weisbach equation. The sign function $\text{sign}(q_{mn}^t)$ indicates the direction of flow in the pipe. Constraint (2) limits the water volume in the pipe. The variable-speed pump is used and modeled by (3) to (5). The pressure reducing valve is adopted, as formulated by (6), (7), and

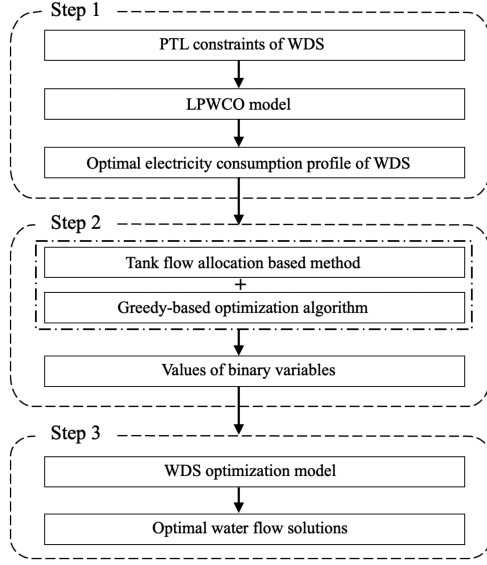


Fig. 1. Three-step optimization procedure.

(8). Equation (9) describes the mass balance, and (10) restricts the pressure head of each node. Constraints from (11) to (14) model the water tank. The electricity consumption of the pump can be obtained by (15). The total electricity consumption of WDS is the sum of all the pumps, as shown in (16).

The sign function, $\text{sign}(q_{mn}^t)$, in the constraint (1) and the ON/OFF status of the pump in constraints (3), (4), and (5) are represented by binary variables using the Big-M approach [3]. The binary variable z_{mn}^t , in (6) and (8) is used to indicate the ON/OFF status of the valve (equals to 1 if on and 0 otherwise). In addition, constraints (1), (3), and (15) are nonlinear. These binary variables and nonlinear constraints make the WDS model become a MINLP problem. The MINLP problem is NP-hard and it may take hours to solve it. The long computation time hinders the WDS from being fully used by power system to deal with the wind power uncertainties. Although the nonlinear constraints can be handled well by many recent researches [24], [25], [26], [27], the binary variable concern in the model remains unresolved. Handling of binary variables in the WDS model is a gap that needs to be filled.

This article aims at tackling the long computation time problem caused by the binary variables in the WDS model. A three-step solution approach is developed to tackle this problem and facilitate the efficient operation. This approach calculates the values of binary variables in advance. The solutions of the WDS model can be obtained very efficiently when binary variable values are known. The efficiency means that the solution time of WDS optimization model is reduced from hours to seconds. This makes it possible for the WDS to adjust its electricity consumption according to the wind power variations so that more wind power can be incorporated into the power system. As a result, the operation cost of the integrated electrical-water system can be significantly reduced.

The proposed three-step approach is shown by the flowchart of Fig. 1. The first step generates the optimal electricity consumption profile of WDS by solving the LPWCO model. The second

step calculates the values of binary variables using the tank flow allocation based method and the greedy-based algorithm. With the values of binary variables, the optimization model of WDS can be solved to obtain the optimal water flow solutions.

III. LPWCO MODEL

This section develops the mathematical formulation of the LPWCO model in Step 1 of the three-step procedure. Prior to the description of the LPWCO model, the PTL constraints of WDS are outlined. These constraints aim at using linear expressions to describe the maximum variation range of the WDS power consumption.

A. PTL Constraints of WDS

The hydraulic constraints from (1) to (16) are nonlinear and involve the binary variables. They should not be violated by the variation of WDS power consumption. These constraints face two shortcomings. One is the computation time is extremely large when incorporated into the real-time operation model of the integrated electrical-water system. Another is that these constraints do not provide an explicit range in which the WDS electricity consumption can vary.

To alleviate the abovementioned two concerns, the PTL constraints were developed to describe the maximum variation range of WDS power consumption. The PTL constraints do not incur much computation burden in real time. They represent the upper and lower bounds of the WDS power consumption as explicit functions of the water volume in the tank, which is critical in modeling the power variation range of WDS.

The variation range of the electrical demand of WDS is affected by constraints (13) and (14). Equation (13) describes the relationship of water volumes in the tank between two consecutive time instants, while (14) imposes restrictions on the water volume in the tank. These two constraints imply that the water volume in the tank at each time instant would have impact on the allowable amounts of tank flow during the next several time periods. For example, when $W_{J_m,t} = W_{J_m}^{\max}$, no more water can be stored by the tank in the next time instant $t + 1$. This may lower the upper bound of the WDS power consumption later, because WDS is prevented from pumping more water into the tank for storage.

Next, we describe the means to develop the PTL constraints. The volume of tank is partitioned into three ranges, as shown in Fig. 2. Fig. 2(a) shows the case when $W_{J_m}^{\min} + q_{J_m}^r \leq W_{J_m}^{\max} - q_{J_m}^r$ and Fig. 2(b) the case when $W_{J_m}^{\min} + q_{J_m}^r > W_{J_m}^{\max} - q_{J_m}^r$. $q_{J_m}^r$ denotes the maximum amount of flows that can be transported in or out of the tank by the inlet and outlet pipes. The size of $W_{J_m}^{\min} + q_{J_m}^r$ and $W_{J_m}^{\max} - q_{J_m}^r$ depends on the technical specifications of the tank.

The partitioning of tank volume can be done by the maximum amount of water that can be stored or released by the tank, which is determined by two factors. One is the value of $q_{J_m}^r$, another is $W_{J_m,t}$. $W_{J_m,t}$ is the water volume in the tank. The value of $q_{J_m}^r$ depends on the technical limitations of the pipe. When the water volume is in the range of $W_{J_m}^{\min} \leq W_{J_m,t} < W_{J_m}^{\min} + q_{J_m}^r$, the maximum amount of water that can be released by the

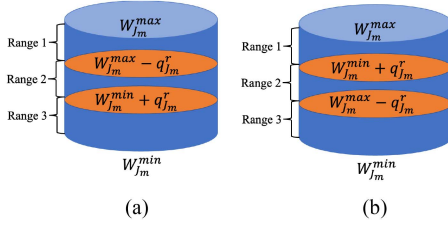


Fig. 2. Tank volume partitioning. (a) $W_{J_m}^{\min} + q_{J_m}^r \leq W_{J_m}^{\max} - q_{J_m}^r$. (b) $W_{J_m}^{\min} + q_{J_m}^r > W_{J_m}^{\max} - q_{J_m}^r$.

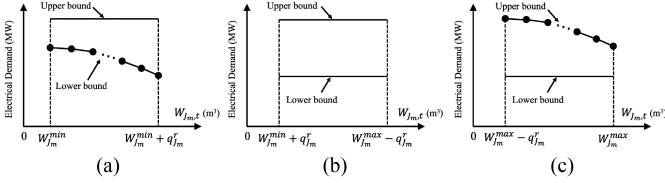


Fig. 3. Maximum variation range of the WDS power consumption. (a) Situation 1. (b) Situation 2. (c) Situation 3.

tank equals $W_{J_m,t} - W_{J_m}^{\min}$, instead of $q_{J_m}^r$. This is due to the fact that the lower bound of the tank storage $W_{J_m}^{\min}$ prevents too much water from being released. Similarly, when $W_{J_m}^{\max} - q_{J_m}^r < W_{J_m,t} \leq W_{J_m}^{\max}$, the maximum amount of water that can be stored by the tank is $W_{J_m}^{\max} - W_{J_m,t}$, obeying the upper bound limitation. It can be seen from Fig. 2(a) that the maximum amount of stored and released water equals $q_{J_m}^r$ and no longer bounded by the tank volume when the value of $W_{J_m,t}$ lies between $W_{J_m}^{\max} - q_{J_m}^r$ and $W_{J_m}^{\min} + q_{J_m}^r$.

In the following, Fig. 2(a) is taken as example to elaborate the process of generating the PTL constraints. This approach also can be applied to the situation of Fig. 2(b). Fig. 2(a) considers three situations of the water volume in the tank as follows:

- Situation 1: $W_{J_m}^{\min} \leq W_{J_m,t} < W_{J_m}^{\min} + q_{J_m}^r$;
- Situation 2: $W_{J_m}^{\min} + q_{J_m}^r \leq W_{J_m,t} \leq W_{J_m}^{\max} - q_{J_m}^r$;
- Situation 3: $W_{J_m}^{\max} - q_{J_m}^r < W_{J_m,t} \leq W_{J_m}^{\max}$.

Fig. 3 shows the maximum variation range of the WDS power consumption under these three situations. The x-coordinate denotes the water volume in the tank and the y-coordinate denotes the electrical demand of WDS. Details for calculating the maximum variation range of the WDS power consumption under each situation are given below.

1) **Situation 1:** WDS increases the electricity consumption by pumping more water into the tank for storage. Since $W_{J_m,t} + q_{J_m}^r < W_{J_m}^{\max}$ in this situation, constraints (13) and (14) are guaranteed to be satisfied when the tank stores water. The upper bound of the WDS power consumption can be calculated as the value of variable $P_{h,t}^{\text{WDS}}$ after solving the optimization model

$$\begin{cases} \max P_{h,t}^{\text{WDS}} \\ \text{s.t. (1)–(12), (15)–(16)} \end{cases} \quad (17)$$

The electrical demand of WDS can be lowered by reducing the electricity consumption of the pump and releasing the water

stored in the tank to satisfy demands. The maximum amount of water that can be released by the tank equals $W_{J_m,t} - W_{J_m}^{\min}$. The minimum electrical demand of WDS can be obtained by solving the optimization model

$$\begin{cases} \min P_{h,t}^{\text{WDS}} \\ \text{s.t. (1)–(16)} \end{cases} \quad (18)$$

when the water volume in the tank equals

$$W_{J_m,t} = W_{J_m}^{\min} + \frac{q_{J_m}^r}{N_D} k, \text{ for } k = 0, 1, 2, \dots, N_D.$$

Each dot in Fig. 3(a) represents the optimization result under a given value of k . These dots are connected by a series of linear segments, which constitutes the lower bound of the WDS power consumption. As N_D increases, the linear segments can approximate the lower bound more accurately, at the cost of increased computation time.

The maximum variation range of the WDS power consumption in Situation 1 can be expressed as

$$f_1(W_{J_m,t}) \leq P_{h,t}^{\text{WDS}} \leq P_{h,t}^{\text{ub}} \quad (19)$$

$f_1(W_{J_m,t})$ is the function describing the lower bound of the WDS power consumption and $P_{h,t}^{\text{ub}}$ is the upper bound.

2) **Situation 2:** Situation 2 ensures that the calculated value of variable $W_{J_m,t+1}$ by (13) obeys the constraint (14). Under this circumstance, constraints (13) and (14) can be removed from the optimization model. The upper bound of the WDS power consumption can be obtained as the value of $P_{h,t}^{\text{WDS}}$ by solving the optimization model (17) and the lower bound by solving the optimization model

$$\begin{cases} \min P_{h,t}^{\text{WDS}} \\ \text{s.t. (1)–(12), (15)–(16)} \end{cases} \quad (20)$$

In this situation, the maximum variation range of the WDS power consumption is

$$P_{h,t}^{\text{lb}} \leq P_{h,t}^{\text{WDS}} \leq P_{h,t}^{\text{ub}} \quad (21)$$

$P_{h,t}^{\text{lb}}$ denotes the value of variable $P_{h,t}^{\text{WDS}}$ after solving the optimization model (20).

3) **Situation 3:** In this situation, $W_{J_m,t} - q_{J_m}^r > W_{J_m}^{\min}$. Thus, constraints (13) and (14) are always satisfied when the tank releases water. The lower bound of the electrical demand of WDS equals the value of $P_{h,t}^{\text{WDS}}$ after solving the optimization model (20).

Fig. 3(c) shows the upper bound of the WDS power consumption, which can be obtained by solving the optimization model (22) and connecting the obtained dots with a series of linear segments

$$\begin{cases} \max P_{h,t}^{\text{WDS}} \\ \text{s.t. (1)–(16)} \\ W_{J_m,t} = W_{J_m}^{\max} - \frac{q_{J_m}^r}{N_D} k \end{cases} \quad \text{for } k = 0, 1, 2, \dots, N_D. \quad (22)$$

The power consumption of WDS in Situation 3 varies in the range of

$$P_{h,t}^{\text{lb}} \leq P_{h,t}^{\text{WDS}} \leq f_3(W_{J_m,t}) \quad (23)$$

$f_3(W_{J_m,t})$ is the function describing the upper bound of the WDS power consumption in Fig. 3(c).

4) PTL Constraints: The PTL constraints consist of the constraints of WDS power variation range and the constraints of tank flow. The former includes (19), (21), and (23). They express the maximum power variation range of WDS as linear functions of the water volume in the tank. The latter is composed of the constraints from (10) to (14). They express the restrictions of water volume and the flows of the tank.

PTL constraints are generated by solving the optimization models (17), (18), (20), and (22). Due to the existence of binary variables and nonlinear constraints, these optimization models are solved in advance to avoid the concern of computation time. The generated PTL constraints are then incorporated into the LPWCO model for efficient real-time operation of the integrated electrical-water system.

$f_1(W_{J_m,t})$ and $f_3(W_{J_m,t})$ in PTL constraints (19) and (23) are linear approximated functions. They approximate the lower and upper bounds of the WDS power consumption. When N_D is set as a fairly large number, $f_1(W_{J_m,t})$ and $f_3(W_{J_m,t})$ act the same as the boundaries of the WDS power consumption. Thus, the PTL constraints bring no loss of optimality to the proposed approach.

B. LPWCO Model

The LPWCO model aims at the cost-effective operation of the integrated electrical-water system in real time. It is solved every 15 min given the updated wind power and electrical demand forecast. The objective function can be expressed as

$$C_A = \min \left(\sum_{h=1}^{N_{WDS}} \sum_{t=t_p}^T \lambda \text{price}_t P_{h,t}^{WDS} + \sum_{w=1}^{N_W} \sum_{t=t_p}^T \lambda P_{w,t}^{wc} \right. \\ \left. + \sum_{n=1}^{N_{EB}} \sum_{t=t_p}^T \mu P_{n,t}^{ec} + \sum_{i=1}^{N_g} \sum_{t=t_p}^T \sum_{l=1}^L K_{i,l} P_{i,t,l}^g \right) \quad (24)$$

λprice_t is the electricity price during period t .

Constraints of the LPWCO model consist of the constraints of the power system and the PTL constraints of WDS. The constraints of the power system are listed as follows:

$$P_{i,t}^g = \sum_{l=1}^L P_{i,t,l}^g \quad \forall i \in \Omega^T \quad (25)$$

$$P_{i,t,l}^g \geq 0 \quad \forall i \in \Omega^T, l \in \Omega^L \quad (26)$$

$$|P_{i,t}^g - P_{i,t-1}^g| \leq \text{Ramp}_i \hat{x}_{i,t-1} + P_i^{g\min} (\hat{x}_{i,t} - \hat{x}_{i,t-1}) \\ + P_i^{g\max} (1 - \hat{x}_{i,t}) \quad \forall i \in \Omega^T, \text{ for } t = t_p + 1, t_p + 2, \dots, T \quad (27)$$

$$|P_{i,t_p}^g - \hat{P}_{i,t_p-1}^g| \leq \text{Ramp}_i \hat{x}_{i,t_p-1} + P_i^{g\min} (\hat{x}_{i,t_p} - \hat{x}_{i,t_p-1}) \\ + P_i^{g\max} (1 - \hat{x}_{i,t_p}) \quad \forall i \in \Omega^T \quad (28)$$

$$P_i^{g\min} \hat{x}_{i,t} \leq P_{i,t}^g \leq P_i^{g\max} \hat{x}_{i,t} \quad \forall i \in \Omega^T \quad (29)$$

$$\sum_{i \in \Omega_n^T} P_{i,t}^g + \sum_{w \in \Omega_n^W} (P_{w,t}^{wr} - P_{w,t}^{wc}) - \sum_{h \in \Omega_n^{WDS}} P_{h,t}^{WDS} - d_{n,t}^{er}$$

$$+ P_{n,t}^{ec} = \sum_{m \in \Lambda_n^E} B_{n,m} (\theta_{n,t} - \theta_{m,t}) \quad \forall n \in \Lambda^E \quad (30)$$

$$0 \leq P_{w,t}^{wc} \leq P_{w,t}^{wr} \quad \forall w \in \Omega^W \quad (31)$$

$$0 \leq P_{n,t}^{ec} \leq d_{n,t}^{er} \quad \forall n \in \Lambda^E \quad (32)$$

$$-PF_{n,m} \leq B_{n,m} (\theta_{n,t} - \theta_{m,t}) \leq PF_{n,m} \quad \forall (n, m) \in \Psi^E \quad (33)$$

$$\theta_{ref,t} = 0. \quad (34)$$

Operation limitations of the unit are described in constraints from (25) to (29). $\hat{x}$ denotes the commitment status of the unit (equals to 1 if on and 0 otherwise), which serves as the input parameter for the optimization model of power system. The value of $\hat{x}$ is determined by solving the day-ahead unit commitment model [17]. Constraint (30) is the nodal flow equation. Constraints from (31) to (32) give the value ranges of the wind power curtailment and electrical load curtailment. Constraint (33) shows the direct current power flow limit of the line. Constraint (34) enforces the phase angle of the reference bus to zero. Solving the LPWCO model yields the value of variable P_{h,t_p}^{WDS} (i.e., $\hat{P}_{h,t_p}^{WDS}$), which denotes the optimal electrical demand of WDS at time t_p .

IV. DETERMINATION OF BINARY VARIABLE VALUES

This section presents the proposed method for determining the values of binary variables in the WDS optimization model. The proposed method includes a tank flow allocation based scheme to find the maximum amount of input and output flows of the tank. Then a greedy-based algorithm is developed to facilitate the computations in finding the appropriate combination of the binary variable values that satisfies the optimal electricity consumption of WDS obtained in Step 1.

A. Tank Flow Allocation Based Method

The tank flow allocation based method allocates the tank flows to the branches in the WDS network. The tank flows are adjusted according to the allocation proportion to overloaded branch. The upper and lower bounds of the WDS power consumption can be obtained by use of this method. We also discuss the difference on the variation ranges of the WDS power consumption obtained from the tank flow allocation based method and the PTL constraints.

1) Tank Flow Allocation: The branch of a WDS denotes the pipe, pump or valve connecting two adjacent nodes. Given the water flows along the branches in the WDS, the matrix R with a size of $N \times N$ is built. The element in the matrix R equals

$$r_{h,m} = \begin{cases} -q_{h,m} & \text{for } h \neq m, q_{h,m} > 0 \\ 0 & \text{for } h \neq m, q_{h,m} \leq 0 \\ \sum_{j \in \Omega_m} q_{j,m} + \sum_{k=1, k \neq m}^N q_{k,m} & \text{for } h = m, q_{k,m} > 0. \end{cases} \quad (35)$$

N is the number of nodes in the WDS. $q_{h,m}$ denotes the amount of water flowing from node n_h to n_m . $q_{h,m} > 0$ when

these two nodes are connected and the direction of flow is from node n_h to n_m ; $q_{h,m} < 0$ when the direction of flow is from n_m to n_h ; $q_{h,m} = 0$ when these two nodes are disconnected. Ω_m denotes the set of tanks connected to node m , and $q_{J_i,m}$ is the input/output flow of tank i . Define E and D as matrices of size $N \times 1$. E is a matrix of ones. d_h denotes the h th element of the matrix D , which is equal to the water demand at node h (equal to 0 if no demand). The relationship between R , E , and D can be expressed as

$$R \cdot E = D. \quad (36)$$

Define W' as a $N \times N$ diagonal matrix with the diagonal element $w'_{m,m} = \sum_{J_i \in \Omega_m} q_{J_i,m}$. Let the matrix W be

$$W = W' \cdot E. \quad (37)$$

It follows from (36) and (37) that

$$W = W' \cdot R^{-1} \cdot D = K \cdot D. \quad (38)$$

Since R is a weakly chained diagonally dominant matrix, it is invertible [28]. $K = W' \cdot R^{-1}$ is a matrix representing the relationship between the input/output flow of the tank and the water demand. Its element $k_{m,h}$ denotes the relationship between the m th tank flow and the demand at node h . Each element in W , w_m , denotes the tank flow at node m and is expressed as $w_m = \sum_{h=1}^N k_{m,h} d_h$. It means that w_m supplies $k_{m,h} d_h$ of the demand at node h .

The mass balance of node h is given as

$$\sum_{k=1, k \neq h}^N q_{k,h} = d_h + \sum_{m=1, m \neq h}^N q_{h,m}. \quad (39)$$

Substituting (39) into $w_m = \sum_{h=1}^N k_{m,h} d_h$ yields

$$w_m = \sum_{h=1}^N \left(k_{m,h} \sum_{k=1, k \neq h}^N q_{k,h} - k_{m,h} \sum_{m=1, m \neq h}^N q_{h,m} \right). \quad (40)$$

It says that the tank flow at node m is equal to the sum of the flow in the branch connected with node h multiply by $k_{m,h}$.

The input and output flows of the tank are allocated to the demand nodes and branches in the WDS network. Larger $k_{m,h}$ means that more demand d_h , and flow connected with node h , $q_{k,h}$, account for the tank flow at node m .

2) Tank Flow Allocation Based Method: The balance of supply and demand in WDS may be threatened when the water flows in the branches violate the maximum volume limitation. To prevent this situation from happening, the input and output flows of the tank should be adjusted according to the flow to overloaded branch. Thus, we develop the tank flow allocation based method to limit the input/output flow of the tank and to find the proper maximum/minimum electrical demand of WDS. Procedure to obtain these limits is given as follows.

- 1) Set the input/output flow of the tank to the maximum threshold satisfying the technical limitations.

- 2) Simulate the operation of WDS using the Newton–Raphson method [29]. Check whether there are overloaded branches. If not, the calculated input/output flow of the tank is the maximum; otherwise, use the tank flow allocation method to calculate the allocation proportion of the input/output flow of the tank to the overloaded branch. Repeat step 2).

For example, for the branch connecting nodes n_h and n_k , reduce the input/output flow of the tank at node m by

$$\frac{(q_{h,k} - q_{h,k}^{\max})}{q_{h,k}} k_{m,h} q_{J_i,m}.$$

$q_{h,k}^{\max}$ is the maximum volume of the water flow.

The Newton–Raphson method is a widely used method for determining the flows in the network [29]. The tank flow allocation based method calculates the water flow solutions by algebraic operations, instead of solving the optimization model. This can reduce the computation time greatly.

3) Discussions: The tank flow allocation based method enables the water flow solutions to be found when the input/output flow of the tank is set to the maximum. Water tank can be regarded as an auxiliary storage device for the power system. More water storage means that more electricity can be consumed by the pump and more wind power can be incorporated into the power system. The upper and lower bounds of the WDS power consumption can be obtained when the input and output flows of the tank are set to the maximum. Equations (15) and (16) are used to calculate the maximum and minimum electrical demands of the WDS.

The PTL constraints provide the possible maximum range that the electricity consumption of WDS can vary. But they did not offer what the variation range of the WDS power consumption is when the binary variable values are known. The maximum variation range of the WDS power consumption under given binary variable values can be obtained by the tank flow allocation method, which is developed based on the given water flow directions. This method also can be used in real time with little computation cost, whereas the PTL constraints need to be generated day ahead.

B. Greedy-Based Algorithm

Algorithm 1 summarizes the solution procedure of the greedy-based algorithm. It contains five steps. The tank flow allocation based method is used in Step 3 to find the power variation range of WDS under the given binary variable values.

The benchmark combinations of the binary variable values in Step 1 can be generated by solving an optimal water flow (OWF) optimization problem as detailed in [25]. The fitness function in Step 4 aims to evaluate the distance between the optimal electricity consumption of WDS and the boundaries of the WDS power variation range. Smaller fitness function value means closer distance. N_V and N_M are hyperparameters of the greedy-based algorithm, and $N_V \times N_M$ represents the number of combinations the algorithm explores each iteration. Larger $N_V \times N_M$ means that more combinations the algorithm can explore. When $N_V \times N_M \geq 2^{N_T}$, it only takes one iteration for the algorithm

Algorithm 1: Greedy-Based Algorithm.

Step 1: Generate N_V combinations of the binary variable values and include them into set $\mathcal{B}$. Initialize the iteration number m as 1.

Step 2: Generate N_M matrices using the Monte Carlo simulation. These matrices are of size $1 \times N_T$, where N_T is the number of binary variables. The elements in the matrix equal 0 or 1 at random. If the j^{th} element in the matrix is equal to 0, then change the value of the j^{th} binary variable. For the i^{th} ($i = 1, 2, \dots, N_V$) combination in the set $\mathcal{B}$, update the binary variable values with the elements in the k^{th} ($k = 1, 2, \dots, N_M$) matrix. If the updated combination has been explored before, resample the k^{th} matrix and update the i^{th} combination.

Step 3: Use the Newton-Raphson method to solve the water flow equations and check whether the updated binary variable values satisfy the hydraulic requirements. If satisfied, calculate the maximum electrical demand, $UB_{h,t_p}^{i,k}$, and the minimum electrical demand, $LB_{h,t_p}^{i,k}$, of the WDS using the tank flow allocation method.

Step 4: Update the i^{th} combination using the k^{th} matrix if the optimal electricity consumption of WDS at time t_p is in the range of $LB_{h,t_p}^{i,k} \leq \hat{P}_{h,t_p}^{WDS} \leq UB_{h,t_p}^{i,k}$, and set the obtained binary variable values as the final solutions. The iterative process is terminated. Otherwise, update the i^{th} combination in the set $\mathcal{B}$ using the matrix with the minimum fitness function value, $fit_i^k = \min\{|\hat{P}_{h,t_p}^{WDS} - LB_{h,t_p}^{i,k}|, |\hat{P}_{h,t_p}^{WDS} - UB_{h,t_p}^{i,k}|\}$.

Step 5: Let $m = m+1$. If $m \leq m_{max}$, go to Step 2; otherwise, update the g^{th} combination using the p^{th} matrix and fix the obtained binary variable values as the final solution

$$fit_g^p = \min\{fit_i^k : i = 1, 2, \dots, N_V; k = 1, 2, \dots, N_M\}$$

m_{max} is the maximum number of iterations.

to converge. 2^{N_T} is equal to the total number of possible combinations.

C. Optimal Operation of the WDS

With the values of binary variables known, the WDS optimization model (41) can be solved to obtain the optimal water flow solutions. The objective function of the WDS optimization model ensures that the optimized operation of WDS satisfies the optimal electricity consumption obtained in Step 1 as much as possible

$$\begin{cases} \min(P_{h,t_p}^{WDS} - \hat{P}_{h,t_p}^{WDS}) \\ \text{s.t. (1)–(16)} \\ P_{h,t_p}^{WDS} \geq \hat{P}_{h,t_p}^{WDS} \end{cases} \quad (41)$$

D. Supplementary Remarks**1) Original Model of Integrated Electrical-Water System:**

The original optimization model of integrated electrical-water system is given as

$$\begin{cases} \min C_A \\ \text{s.t. (1)–(16)} \\ (25)–(34) \end{cases} \quad (42)$$

The detailed formulation of C_A can be seen from (24), which aims at minimizing the operation cost of the integrated electrical-water system. The proposed three-step approach decomposes the solution process of the MINLP model (42) into three steps, which enables the efficient operation of the integrated electrical-water system.

2) Optimality of Greedy-Based Algorithm: The proposed greedy-based algorithm can guarantee that the global optimal solution is found. This is explained as follows.

The greedy-based algorithm aims at finding the optimal combination of the binary variable values. Since there are limited numbers of possible combinations, the global optimal one is guaranteed to be found when $m_{max} \geq \frac{2^{N_T}}{N_V \times N_M}$ in Step 5. At each iteration, the greedy-based algorithm calculates the range that the electrical demand of WDS can vary for each updated combination. The combination is determined as the global optimal one when the optimal electricity consumption of WDS is in the calculated range. This greatly accelerates the convergence of the greedy-based algorithm since there are many combinations of binary variable values that satisfy $LB_{h,t_p}^{i,k} \leq \hat{P}_{h,t_p}^{WDS} \leq UB_{h,t_p}^{i,k}$. Under this situation, any combination can be chosen as the final solution without affecting the optimality in that the optimal electricity consumption of WDS is satisfied under these combinations.

3) Time Complexity Analysis: The big O notation is used to analyze the time complexity of the proposed three-step solution process. The time complexity is calculated by

$$T(n) = T(n_1) + T(n_2) + T(n_3) \quad (43)$$

$T(n_1)$, $T(n_2)$, and $T(n_3)$ denote the time complexity of Steps 1, 2, and 3, respectively. Since the PTL constraints are generated in advance, the time complexity of Step 1 equals the time complexity for solving the LPWCO model. $T(n_3)$ also equals the time complexity for solving the linear programming model (41). According to [30], we have

$$T(n_1) = T(n_3) = O(n^2). \quad (44)$$

The time complexity of Step 2 is calculated by

$$\begin{aligned} T(n_2) &= \sum_{k=1}^5 T(S_k) = 1 + n + O(\max(n^3, n^3)) \\ &+ n + 1 = O(n^3) \end{aligned} \quad (45)$$

$T(S_k)$ denotes the time complexity of Step k in the greedy-based algorithm. Substituting (44) and (45) into (43) yields

$$T(n) = O(\max(n^2, n^3, n^2)) = O(n^3). \quad (46)$$

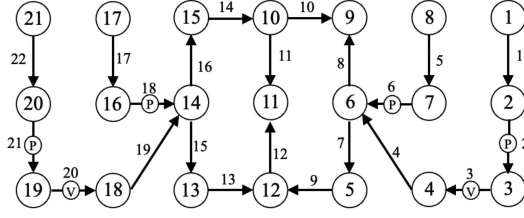


Fig. 4. WDS network comprising 21 nodes (the circles) and 22 branches. The arrows denote the direction of flow in the branches. P denotes the pump and V the valve. Demand loads are at nodes 9, 10, 11, 12, and 15. The four reservoirs are located in nodes 1, 8, 17, and 21. Two tanks are situated at nodes 4 and 18.

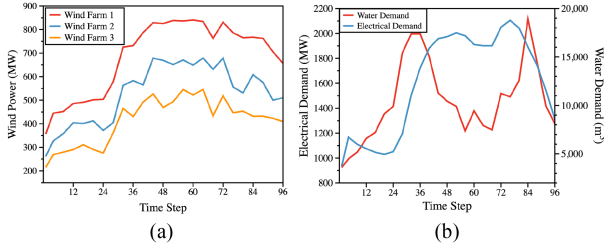


Fig. 5. Daily wind power output and load demands.

The abovementioned analysis shows that the NP-hard optimization problem (42) is turned into a polynomial-time problem by use of the proposed three-step solution approach. This verifies the advantage of the proposed approach in reducing the computational cost.

V. CASE STUDY

The proposed approach for efficient operation of integrated electrical-water system is evaluated to show its merits: much lower wind power curtailment and significantly shorter computation time. This makes it better equipped to deal with the uncertainty of wind power. Performance evaluation was carried out on a small-scale illustrative test system and a large-scale test system. In this article, the time horizon of each step is 24 h. The resolution is 15 min, corresponding to 96 time steps. The tests were performed on a PC with Intel Core i7-9700 and Gurobi 9.0.1 on Python 3.7. This article considers the WDS connected to the power transmission system to evaluate the impact of WDS optimization on the entire power transmission system [17].

A. Illustrative Test System

Firstly, a relatively small test system is used to evaluate the performance of the proposed approach in detail. The test system is composed of the IEEE RTS 24-bus power system and the WDS with 22 branches [7]. The diagram of IEEE RTS 24-bus system can be found in [8]. Fig. 4 shows the test WDS. There are 17 load buses in the power system and each load is connected with a WDS. The wind farms are located at nodes 3, 5, and 20 of the power system. Fig. 5(a) illustrates the daily wind power outputs for three different wind farms and Fig. 5(b) the daily electrical and water load demands. Parameters of the pump are given in

TABLE I
PARAMETER VALUES IN HYDRAULIC EQUATIONS OF PUMP

Parameter	a_{p_m}	b_{p_m}	c_{p_m}	$\hat{\omega}_{p_m}$
Value	-1.0941×10^{-4}	0.5152	40	10

TABLE II
ELECTRICITY PRICE OF WDS

	Valley	Off-peak	Peak
Time period (h)	1–10	11–17 and 23–24	18–22
Price (\$/MWh)	40	70	132

TABLE III
OPTIMIZATION RESULTS UNDER DIFFERENT CASES IN THE ILLUSTRATIVE TEST SYSTEM

Case	Wind Power Curtailment (MWh)	Operation Cost (\$)	Computation Time (sec)
1	1023.42	1057707	2.93
2	735.66	758562	2311.59
3	735.66	758562	2046.27
4	374.06	396721	25.81

Table I. The electricity price for the WDS is presented in Table II. The penalty for wind power curtailment is 1000 \$/MWh. In the greedy-based algorithm $N_V = 50$ and $N_M = 100$.

Four cases are considered to demonstrate the advantage of the proposed optimization approach.

- Case 1: Power system and WDS operate independently.
- Case 2: Cooperative operation of power system and WDS not considering the handling of binary variables. The cooperative operation model is solved in a centralized way.
- Case 3: Cooperative operation of power system and WDS not considering the handling of binary variables. The cooperative operation model is solved by the Benders decomposition approach.
- Case 4: Cooperative operation of power system and WDS considering the handling of binary variables with the proposed three-step optimization approach.

In Case 1, the optimization models of power system and WDS are formulated as (47) and (48), respectively. The optimization models of Cases 2 and 3 are given by (42). Note that Case 2 solves the optimization model (42) directly, whereas Case 3 decomposes the original model by use of the Benders decomposition. The process of Benders decomposition can be seen from [8]

$$\left\{ \begin{array}{l} \min \left(\sum_{w=1}^{N_W} \sum_{t=t_p}^T \lambda P_{w,t}^{wc} + \sum_{n=1}^{N_{EB}} \sum_{t=t_p}^T \mu P_{n,t}^{ec} \right. \\ \left. + \sum_{i=1}^{N_g} \sum_{t=t_p}^T \sum_{l=1}^L K_{i,l} P_{i,t,l}^g \right) \\ \text{s.t. (25)–(34)} \end{array} \right. \quad (47)$$

$$\left\{ \begin{array}{l} \min \sum_{h=1}^{N_{WDS}} \sum_{t=t_p}^T \lambda \text{price}_t P_{h,t}^{WDS} \\ \text{s.t. (1)–(16)} \end{array} \right. \quad (48)$$

Table III shows the optimization results under different cases. The time for solving the optimization model in different cases

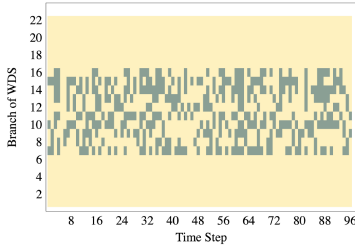


Fig. 6. Binary variable values denoting the direction of flow in the 22 branches over 96 time steps. When the direction of flow in the branch is the same as that of Fig. 4, the binary variable has a value of 1, represented by yellow. Otherwise, the binary variable has a value of 0 and represented by green.

is shown in the fourth column. It is observed from Table III that Case 1 requires the largest wind power curtailment and operation cost. This indicates that the integration of WDS energy flexibility into the power system operation greatly promotes the wind power accommodation. It is also observed from Table III that the amount of wind power curtailment and the operation cost in Case 4 are significantly lower than that in Cases 2 and 3. They are less, respectively, by about 55% and 53%. This is because, in Cases 2 and 3, it takes more than 15 min to solve the WDS optimization problem. Both the centralized optimization and the Benders decomposition approaches are not designed to handle the NP-hard problem caused by the binary variables. Under this circumstance, an immediate response according to the real-time variations of wind power is not available to deal with the wind power uncertainties. Considering the handling of binary variables, the computation time of Case 4 is reduced by 99%. Since the computation time in Case 4 is short enough to adjust the operation strategy of WDS, proper response to the wind power variations could be provided. As a result, wind power curtailment and the operation cost are lowered.

The greedy-based algorithm finds the 96 combinations of binary variable values. Each combination shows the branch status corresponding to a time step in the daily operation. Fig. 6 shows the obtained binary variable values. It can be seen from Fig. 6 that the directions of flow in branches 1 to 6 and 17 to 22 remain fixed. This is because these branches are not within the loop and the water flow direction is generally from the water source to the demand load.

Monte Carlo simulation was applied to generate 500 wind power output profiles based on the scaled data of the Irish grid [8]. Under each profile, we compare the optimization results in Case 4 with the results in Cases 2 and 3 to further test the efficiency of the proposed approach. Fig. 7(a) shows the computation time of Case 4 versus the time in Cases 2 and 3 for 500 profiles. Each dot represents the optimization result under a profile. Note that the time scale for Case 4 is in the magnitude of seconds and that for Cases 2 and 3 in minutes. Fig. 7(b) depicts the percentage difference between the optimized wind power curtailment of Case 4 and that of the value in other cases versus the difference between the operation costs. Note that the optimization results in Cases 2 and 3 are the same. It can be seen from Fig. 7(b) that the proposed approach reduces the wind

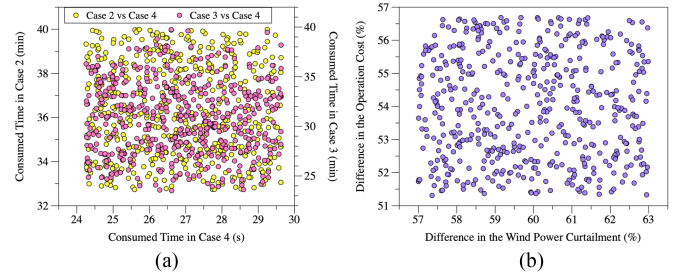


Fig. 7. Comparison of optimization results for 500 generated wind power output profiles.

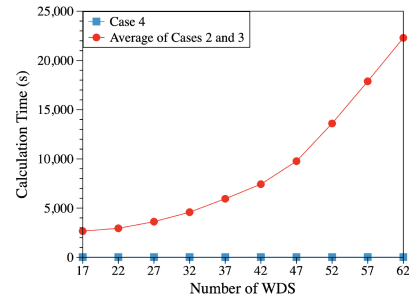


Fig. 8. Impact of the number of WDS on the computation time of the illustrative test system.

power curtailment and the operation cost by about 60% and 54%, respectively. Results in Fig. 7 demonstrate that the proposed approach is computationally efficient for WDS optimization problem. The gap between the optimization results in Case 4 and the global optimal solution is narrowed down to 0 under the 500 wind power profiles. This indicates that the proposed approach is highly accurate.

As the number of WDS connected to the power system increases, the binary variables indicating the direction of flow in the pipe and the ON/OFF status of the pump and valve also increase. Fig. 8 shows the impact of the number of WDS on the computation time. It can be seen from Fig. 8 that, as the number of WDS grows, the average computation time to solve the optimization model in Cases 2 and 3 increases dramatically, while that in Case 4 remains small and almost unchanged. This is all because of the handling of binary variables in the proposed three-step optimization approach. Note that the computation time in Case 4 is in a few seconds, which fully meets the needs of dispatch in real time.

B. Large-Scale Test System

A realistic power system in a province of Northwest China is used to further verify the superiority of the proposed approach. The power system has 93 load buses and 179 transmission lines. The installed wind power capacity is 3667.2 MW, accounting for 12% of the total generating capacity. The “D-Town” WDS with 443 pipes is used [8], which often serves as the benchmark for WDS optimization models. Each load bus in the power system is connected with a WDS.

Consider the same four cases as in the illustrative test system study. In each case, four scenarios are considered for seasonal

TABLE IV
OPTIMIZATION RESULTS UNDER DIFFERENT CASES IN THE LARGE-SCALE TEST SYSTEM

Case	Scenario	Wind Power Curtailment (MWh)	Operation Cost (\$)	Computation Time (sec)
1	<i>a</i>	11763.95	13831583	8.25
	<i>b</i>	8619.49	9889072	7.93
	<i>c</i>	5337.88	6352357	9.27
	<i>d</i>	1641.53	1967376	8.64
2	<i>a</i>	5250.75	6058920	8709.30
	<i>b</i>	2894.93	3259455	8324.69
	<i>c</i>	1896.02	2260780	8637.41
	<i>d</i>	636.35	677753	8409.39
3	<i>a</i>	5250.75	6058920	7239.34
	<i>b</i>	2894.93	3259455	7203.67
	<i>c</i>	1896.02	2260780	6655.15
	<i>d</i>	636.35	677753	6508.80
4	<i>a</i>	2538.71	2707205	64.74
	<i>b</i>	1419.07	1559896	61.80
	<i>c</i>	1005.25	1130063	59.25
	<i>d</i>	0	113814	60.33

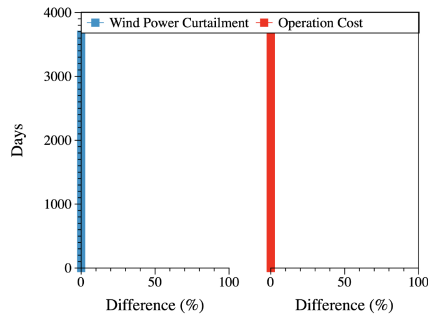


Fig. 9. Comparison of optimization results for ten-year historical scenarios.

impact. Scenario *a* denotes the months from March to May, Scenario *b* from June to August, Scenario *c* from September to November, and Scenario *d* from December to February. Table IV shows the optimization results on wind power curtailment, operation cost, and computation time under these four cases. It can be seen from Table IV that the amount of wind power curtailment and the operation cost are much lower in Case 4 compared with those in other cases. Besides, the computation time is significantly reduced by nearly 99% with the consideration of binary variable handling in WDS model.

Ten-year historical scenarios of the provincial power system were used to further test the optimality of the proposed approach. We compare the optimization results in Case 4 and the global optimal solutions, as shown in Fig. 9. It can be seen from Fig. 9 that the proposed approach can always find the global optimal solution, which verifies the optimality of the proposed approach.

Next, the number of WDS connected to the power system was set to vary from 93 to 203. Fig. 10 shows the impact of the number of WDS on computation time. It is observed from Fig. 10 that the computation time for solving the model in Case 4 remains around 1 min as the number of WDS increases, whereas that in Cases 2 and 3 increases dramatically. This demonstrates that the proposed method can be applied to much larger systems with little increase on the computational cost.

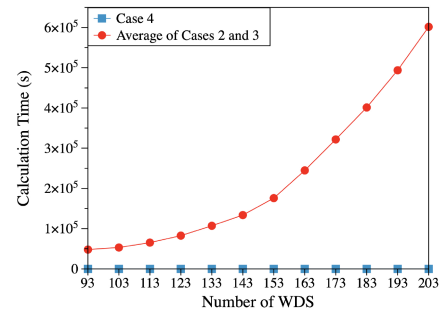


Fig. 10. Impact of the number of WDS on the computation time of the large-scale test system.

VI. CONCLUSION

This article presented a three-step optimization approach for efficient operation of integrated electrical-water system. The efficiency is in terms of significantly shorter computation time and much lower wind power curtailment. The first step provide several linear constraints to model the maximum variation range of the electrical demand of WDS. The optimal electricity consumption profile of WDS was obtained by solving the LPWCO model. The binary variable values of WDS were determined in the second step through the tank flow allocation based method and the greedy-based algorithm. Then, the optimization model of WDS can be solved efficiently. Data privacy can also be protected, since only limited information needs to be shared between WDS and the power system.

Results from case studies demonstrated that the proposed method offers significant benefits on lowering the computation time and the operation cost in comparison to the previous studies. The computation time remains nearly unchanged even as the number of WDS integrated with the power system increases from 93 to 203.

REFERENCES

- [1] International Energy Agency, "World energy outlook 2020." [Online]. Available: <https://www.iea.org/reports/world-energy-outlook-2020>
- [2] M. Khodayar, O. Kaynak, and M. E. Khodayar, "Rough deep neural architecture for short-term wind speed forecasting," *IEEE Trans. Ind. Inform.*, vol. 13, no. 6, pp. 2770–2779, Dec. 2017.
- [3] H. Wei, X. Liu, W. Cao, G. Ye, X. Jiang, and Y. He, "Ultra-short-term wind power forecasting based on deep belief network and wavelet denoising," in *Proc. IEEE 4th Int. Elect. Energy Conf.*, 2021, pp. 1–6.
- [4] T. Yan, W. Liu, M. Zhao, Y. Liu, W. Yu, and L. Long, "Multi-state probability prediction of wind power based on deep learning model," in *Proc. IEEE Sustain. Power Energy Conf.*, 2020, pp. 155–160.
- [5] E. S. Miele, F. Bonacina, and A. Corsini, "Deep anomaly detection in horizontal axis wind turbines using graph convolutional auto-encoders for multivariate time series," *Energy AI*, vol. 8, pp. 1–13, 2022.
- [6] International Energy Agency, "World energy outlook 2018." [Online]. Available: <https://www.iea.org/reports/world-energy-outlook-2018>
- [7] C. Li, Y. Yao, K. Xie, B. Hu, and T. Niu, "Integrated electrical, heating, and water distribution system to accommodate wind power," *IEEE Trans. Sustain. Energy*, vol. 12, no. 2, pp. 1100–1114, Apr. 2021.
- [8] Y. Yao, C. Li, K. Xie, H.-M. Tai, B. Hu, and T. Niu, "Optimal design of water tank size for power system flexibility and water quality," *IEEE Trans. Power Syst.*, vol. 36, no. 6, pp. 5874–5888, Nov. 2021.
- [9] D. Fooladivanda, A. D. Domínguez-García, and P. W. Sauer, "Utilization of water supply networks for harvesting renewable energy," *IEEE Control Netw. Syst.*, vol. 6, no. 2, pp. 763–774, Jun. 2019.

- [10] C. Ashley, L. Holmes, and G. Wikler, "Using more energy can be a good thing: C&I loads as a balancing resource for intermittent renewable energy," EnerNOC Utility Solutions, White Paper, 2013.
- [11] K. Smith, "Understanding energy consumption for urban water supply in China," M.S. thesis, Dept. Environ. Sci. Eng., Tsinghua Univ., Beijing, China, 2015.
- [12] World Bank, "World development indicators," 2016.
- [13] S. Zuloaga and V. Vittal, "Integrated electric power/water distribution system modeling and control under extreme mega drought scenarios," *IEEE Trans. Power Syst.*, vol. 36, no. 1, pp. 474–484, Jan. 2021.
- [14] C. D'Ambrosio, A. Lodi, S. Wiese, and C. Bragalli, "Mathematical programming techniques in water network optimization," *Eur. J. Oper. Res.*, vol. 243, no. 3, pp. 774–788, 2015.
- [15] H. Mala-Jetmarova, N. Sultanova, and D. Savic, "Lost in optimisation of water distribution systems? A literature review of system operation," *Environ. Model. Softw.*, vol. 93, pp. 209–254, 2017.
- [16] W. N. Lubega and A. M. Farid, "Quantitative engineering systems modeling and analysis of the energy–water nexus," *Appl. Energy*, vol. 135, pp. 142–157, 2014.
- [17] K. Oikonomou and M. Parvania, "Optimal coordination of water distribution energy flexibility with power systems operation," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 1101–1110, Jan. 2019.
- [18] K. Oikonomou and M. Parvania, "Optimal coordinated operation of interdependent power and water distribution systems," *IEEE Trans. Smart Grid*, vol. 11, no. 6, pp. 4784–4794, Nov. 2020.
- [19] K. Oikonomou, M. Parvania, and R. Khatami, "Optimal demand response scheduling for water distribution systems," *IEEE Trans. Ind. Inform.*, vol. 14, no. 11, pp. 5112–5122, Nov. 2018.
- [20] K. Oikonomou and M. Parvania, "Optimal participation of water desalination plants in electricity demand response and regulation markets," *IEEE Syst. J.*, vol. 14, no. 3, pp. 3729–3739, Sep. 2020.
- [21] R. Menke, E. Abraham, P. Parpas, and I. Stoianov, "Demonstrating demand response from water distribution system through pump scheduling," *Appl. Energy*, vol. 170, pp. 377–387, 2016.
- [22] H. Shuai, J. Fang, X. Ai, J. Wen, and H. He, "Optimal real-time operation strategy for microgrid: An ADP-based stochastic nonlinear optimization approach," *IEEE Trans. Sustain. Energy*, vol. 10, no. 2, pp. 931–942, Apr. 2019.
- [23] S. Leyffer and J. Lee, *Mixed Integer Nonlinear Programming*. Berlin, Germany: Springer, 2012.
- [24] D. Verleye and E. Aghezaf, "Optimising production and distribution operations in large water supply networks: A piecewise linear optimisation approach," *Int. J. Prod. Res.*, vol. 51, no. 23/24, pp. 7170–7189, 2013.
- [25] D. Fooladivanda and J. A. Taylor, "Energy-optimal pump scheduling and water flow," *IEEE Control Netw. Syst.*, vol. 5, no. 3, pp. 1016–1026, Sep. 2018.
- [26] A. S. Zamzam, E. Dall'Anese, C. Zhao, J. A. Taylor, and N. D. Sidiropoulos, "Optimal water-power flow problem: Formulation and distributed optimal solution," *IEEE Control Netw. Syst.*, vol. 6, no. 1, pp. 37–47, Mar. 2019.
- [27] Q. Li, S. Yu, A. S. Al-Sumaiti, and K. Turitsyn, "Micro water-energy nexus: Optimal demand-side management and quasi-convex hull relaxation," *IEEE Control Netw. Syst.*, vol. 6, no. 4, pp. 1313–1322, Dec. 2019.
- [28] A. Enshaee, G. R. Yousefi, and A. Ebrahimi, "Allocation of transmission active losses through a novel power tracing-based technique," *IET Gener., Transmiss. Distrib.*, vol. 12, no. 13, pp. 3201–3211, 2018.
- [29] W. Viessmand Jr. et al., *Water Supply and Pollution Control*, 8th ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 2008.
- [30] M. B. Cohen, Y. T. Lee, and Z. Song, "Solving linear programs in the current matrix multiplication time," *J. ACM*, vol. 68, no. 1, pp. 1–39, 2021.



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